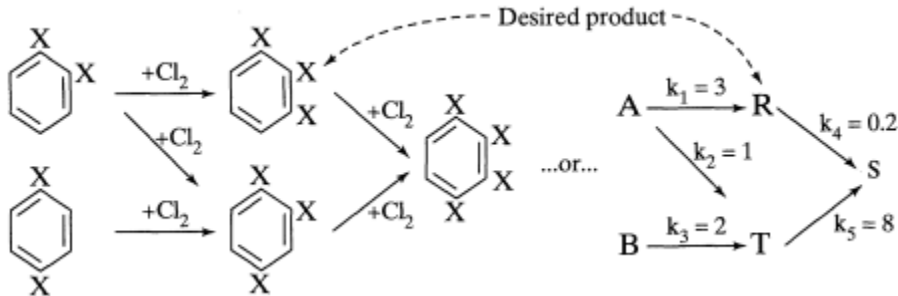


Reaction Engineering II- Individual Assignment (10%)

Submission date: 03/08/2016 E.C

1. The progressive chlorination of o- and p-dichlorobenzene proceed with second-order rate, as shown below.



For a single-feed stream having $C_{A0} = 2$, $C_{B0} = 1$, and 1,2,3-trichlorobenzene as the desired product

- (a) Tell which kind of flow reactor is best.
 (b) In this reactor, find $C_{R,max}$
2. The elementary irreversible organic liquid-phase reaction: $A + B \rightarrow C$ is carried out adiabatically in a flow reactor. An equal molar feed in A and B enters at 27°C, and the volumetric flow rate is 2 dm³/s.

Additional information:

$$H_A^\circ(273) = -20 \text{ kcal/mol}$$

$$H_B^\circ(273) = -15 \text{ kcal/mol}$$

$$H_C^\circ(273) = -41 \text{ kcal/mol}$$

$$C_{A0} = 0.1 \text{ kmol/m}^3$$

$$C_{pA} = C_{pB} = 15 \text{ cal/mol} \cdot \text{K} \quad C_{pC} = 30 \text{ cal/mol} \cdot \text{K}$$

$$k = 0.01 \frac{\text{dm}^3}{\text{mol} \cdot \text{s}} \text{ at } 300 \text{ K} \quad E = 10,000 \text{ cal/mol}$$

- a) Calculate the PFR and CSTR volumes necessary to achieve 85% conversion.
 b) What is the maximum inlet temperature one could have so that the boiling point of the liquid (550K) would not be exceeded even for complete conversion?